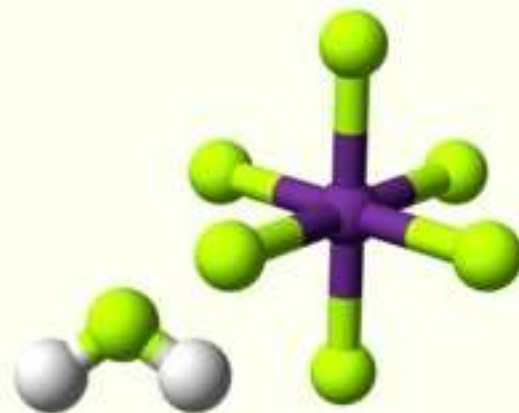


CHEMISTRY

ACIDS, BASES AND SALTS

MODULE - 9



WHY DO FOODS HAVE DIFFERENT TASTE?



SOUR IN
TASTE



BITTER IN
TASTE



Foods taste sour or bitter due to the presence of some compounds in them which give them the particular taste



COMPOUNDS

ACIDS



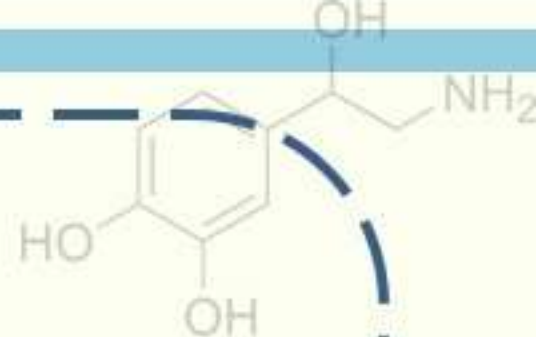
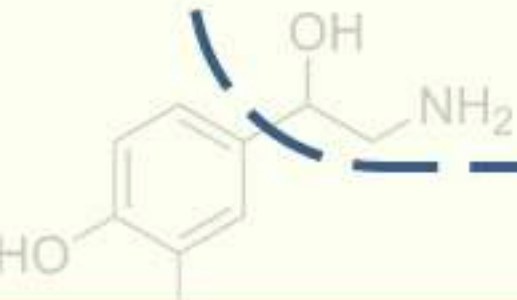
BASES



SALTS



Acids bases and salts are made up two or more than two elements in fixed proportion



CONTRASTING FEATURES

ACIDS

- An acid is a substance that dissociates **in water** to yield H^+ ions.
- The H^+ ions react with water to produce H_3O^+ ions.
- Because of the H^+ ions they taste sour.

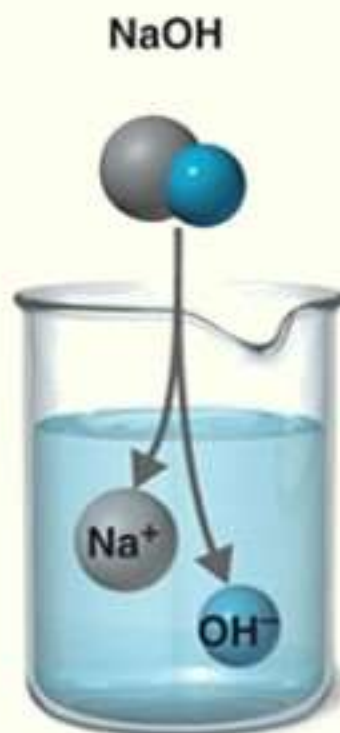
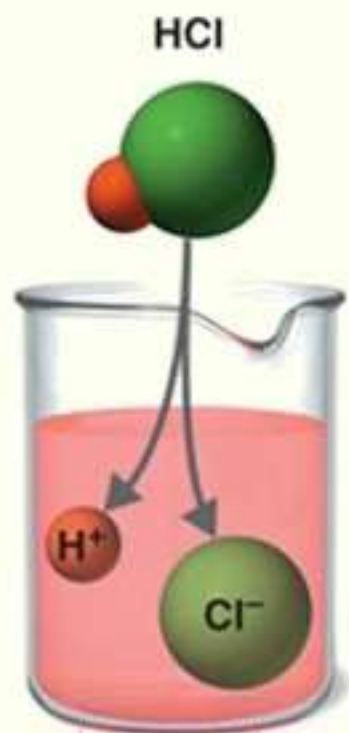
BASES

- A base is a substance that dissociates **in water** to yield OH^- ions.
- The OH^- ions are a natural part of water (self ionisation)
- Because of the OH^- ions they taste bitter.

H_3O^+ and OH^- ions together produce back the water molecule at equilibrium conditions



WILL AN ACID AND A BASE BEHAVE THE SAME WAY THEY DO WITHOUT WATER ?



- An acid and a base will only show their acidic and basic properties when dissolved in water.
- This is because the ions only dissociate in water and they remain separate in it, thus able to react with other matter.
- The ions are capable of conducting electricity only when they are dissociated.

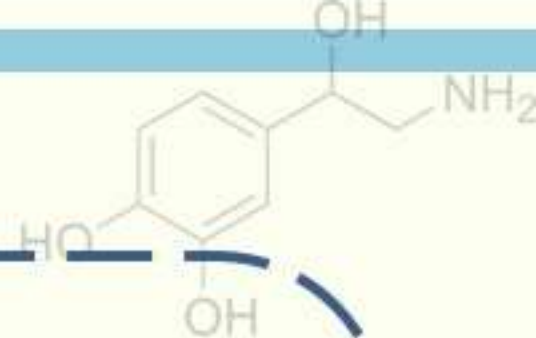
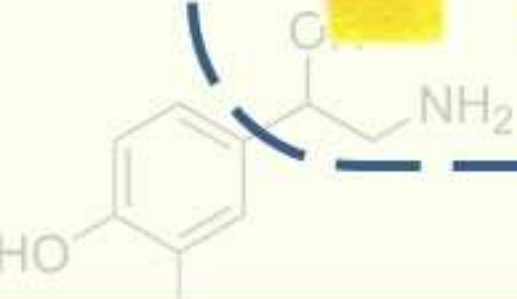
Bases soluble in water are called alkali.

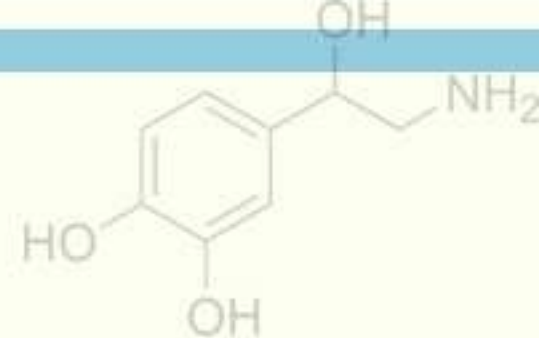


ACID BASE INDICATORS



- Some substances generally change to different colours in the presence of acid or bases respectively. They are called indicators.
- Some other substances can change odour too.
- They can be naturally obtained or scientifically prepared.





ACID BASE INDICATORS

NATURAL

SYNTHETIC

COLOUR CHANGE

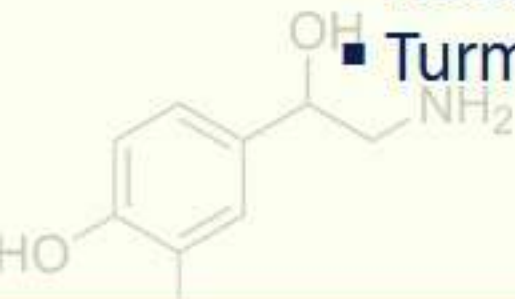
- Litmus
- Beetroot
- Rose petals
- Turmeric

ODOUR CHANGE

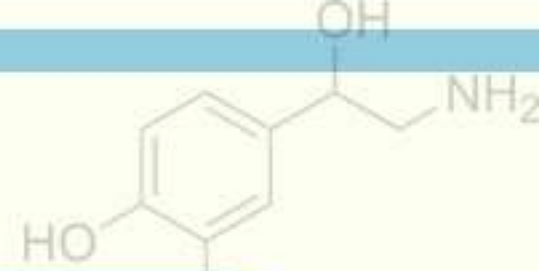
- Vanilla
- Onions

Phenolphthalein

Methyl orange



COLOUR CHANGE IN INDICATORS



ACID

LITMUS



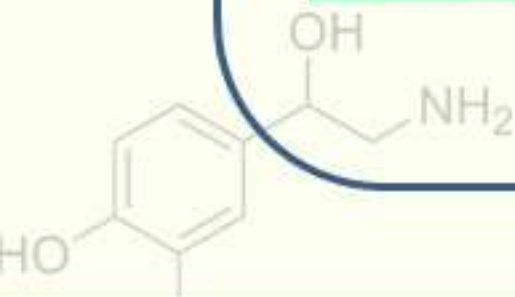
METHYL ORANGE



PHENOLPHTHALEIN



BASE

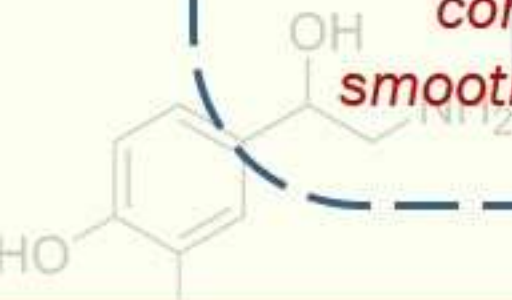
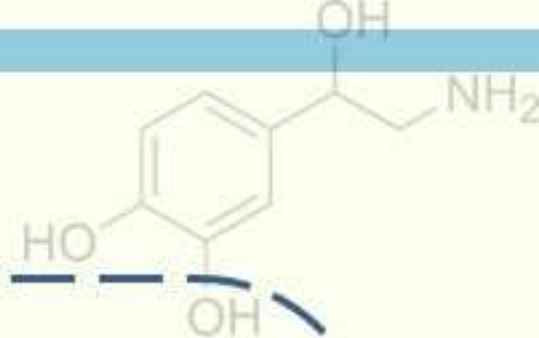


UNIVERSAL INDICATOR

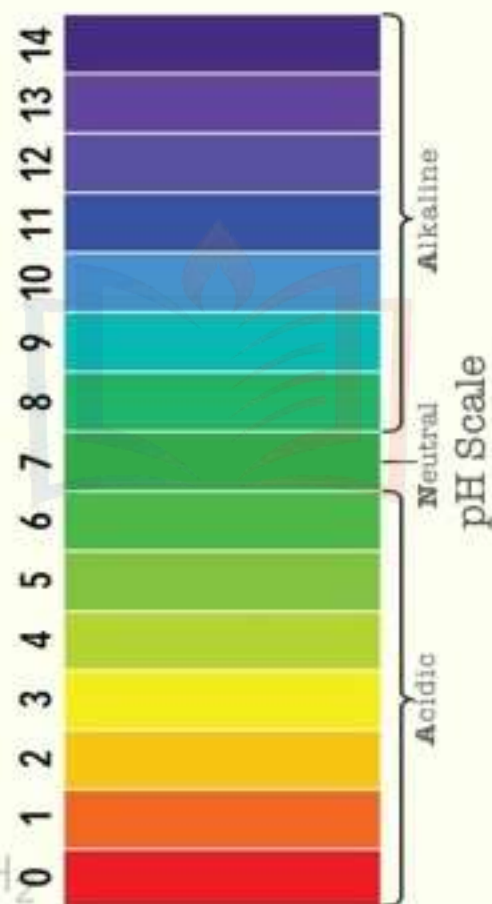


A universal indicator is a pH indicator composed of a solution of several compounds that exhibits several smooth colour changes over a pH value

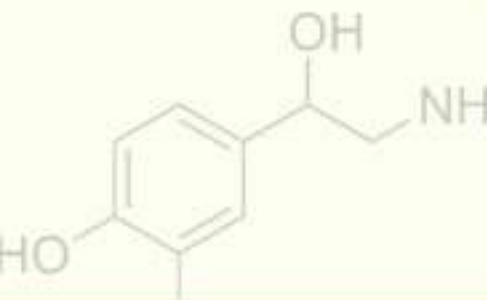
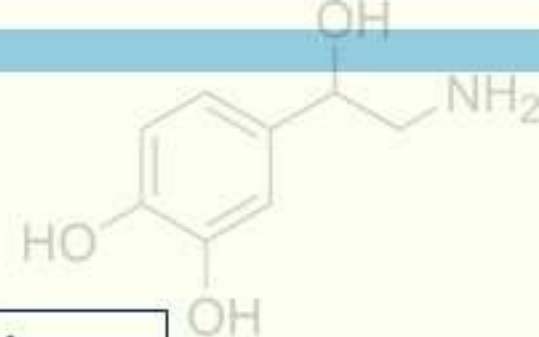
- **Green** colour for neutral solution,
- **Red** for extremely acidic
- **Violet** or **indigo** for basic solutions.
- **Yellow** or **orange** for a weak acid.
- **Blue** colour denotes some weak bases.



THE CONCEPT OF pH VALUE



- pH or potenz(power) of Hydrogen is a scale to show how much hydrogen ions is present in any solution.
- The pH scale is a numeric scale which denotes how much acidic and basic any solution is.
- pH 7 denotes neutral solution i.e. distilled water.
- pH less than 7 denotes an acidic solution.
- pH more than 7 denotes a basic solution.



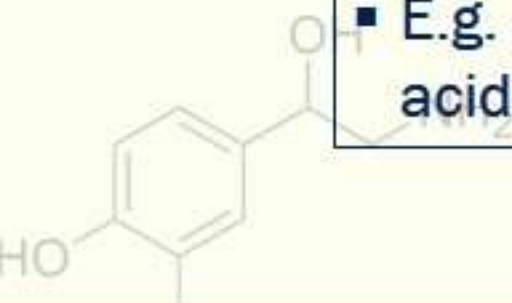
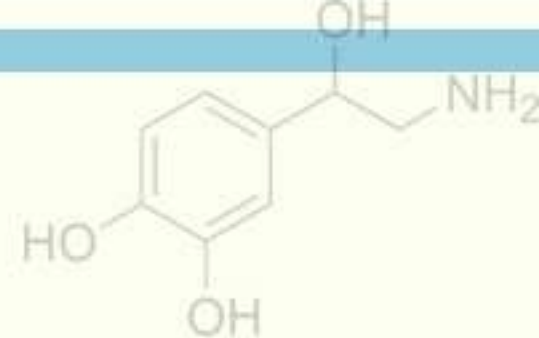
TYPES OF ACIDS

STRONG ACIDS

- Complete dissociation into H^+ ions in water
- Releases all the H^+ ions to the solution.
- pH is very low within 0 to 3.
- Strong acids are very corrosive in nature.
- E.g. hydrochloric acid, nitric acid, sulphuric acid

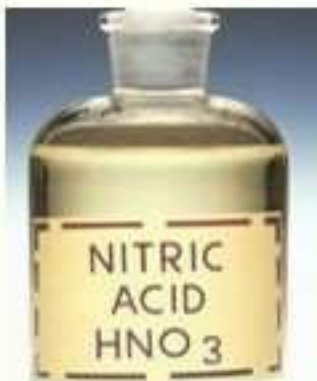
WEAK ACIDS

- Partial dissociation into H^+ ions in water
- Do not release all the H^+ ions to the solution.
- pH is moderate about 3 - 5
- Weak acids are less corrosive in nature.
- E.g. Acetic acid, lactic acid, tartaric acid

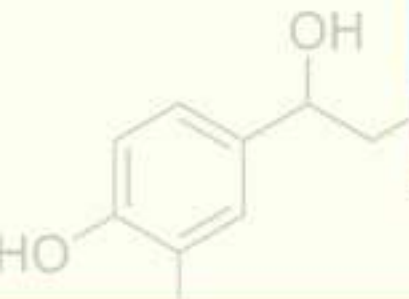
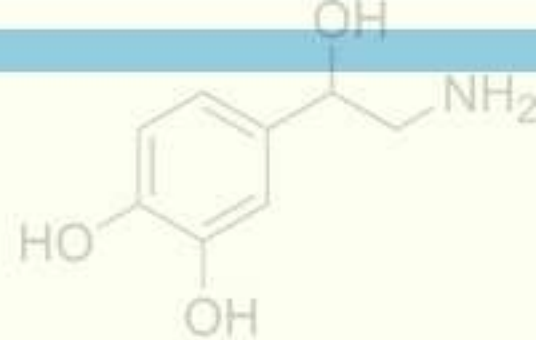


TYPES OF ACIDS

STRONG ACIDS



WEAK ACIDS



TYPES OF BASES

STRONG BASE

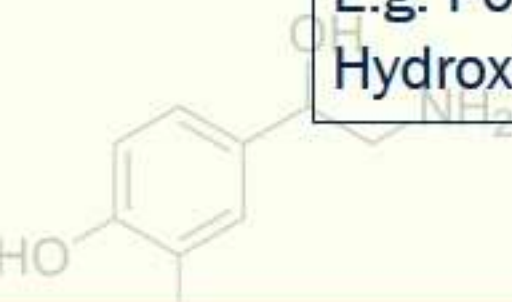
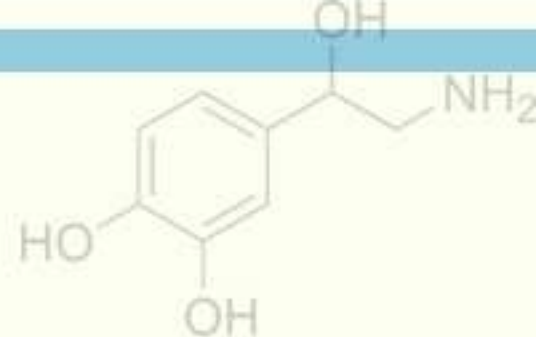
- Complete dissociation into OH^- ions in water
- Releases all the OH^- ions to the solution.
- pH is very high within 10 to 14.
- Strong bases are very corrosive in nature.

E.g. Potassium hydroxide, sodium Hydroxide, calcium hydroxide

WEAK BASE

- Partial dissociation into OH^- ions in water
- Do not release all the OH^- ions to the solution.
- pH is moderate; from 7 to 8
- Weak bases are less corrosive in nature.

E.g. ammonium hydroxide, zinc hydroxide, hydrazine,



TYPES OF BASES

STRONG BASE



WEAK BASE



ORGANIC SOURCES OF ACIDS



ACETIC ACID	NATURAL SOURCES
Acetic acid	Vinegar
Formic acid	Citrus fruits, bee sting
Lactic acid	Curd
Oxalic acid	Spinach
Ascorbic acid	Amla, citrus fruits
Tartaric acid	Tamarind, grapes
Maleic acid	Apples, pears, grapes
Oleic Acid	Olive oil
Folic acid	Lettuce leaves, spinach
Stearic acid	Mustard oil, rape seed oil

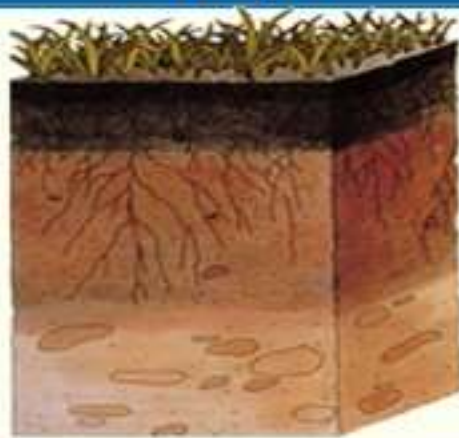


SOME COMMON APPLICATIONS OF ACIDS AND BASES

ORAL HYGEINE



SOIL pH



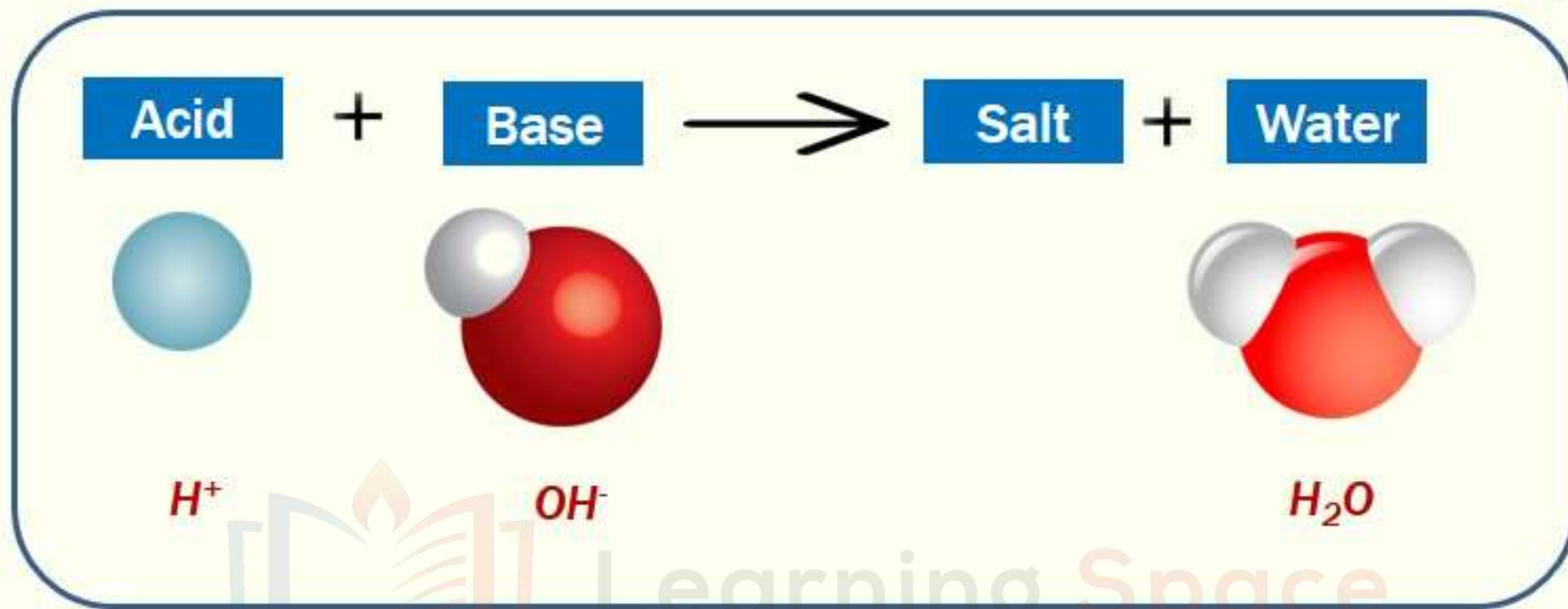
STOMACH ACIDITY



- Our oral cavity or mouth has a pH 7. But after we take food, the bacterial growth slowly decreases the pH to 5.5 – 6.
- A pH of 5.5 to 7.5 of soil is essential for many plants. Acidity and alkalinity of soil may affect the crop yield.
- Our stomach produces very corrosive HCl of low pH during digestion.



NEUTRALIZATION REACTION



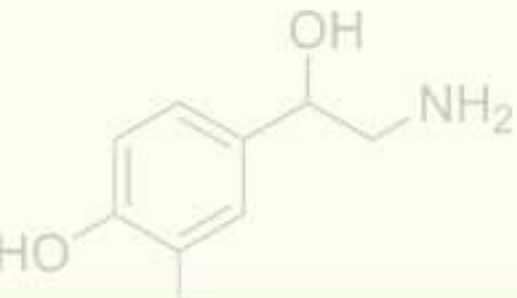
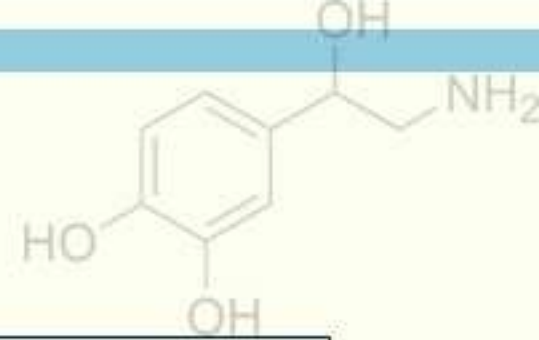
Salt is made by the negative ion of the acid and the positive ion of the base.

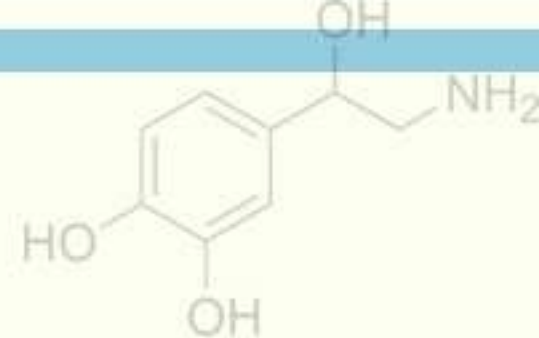


SALIENT FEATURES OF SALTS

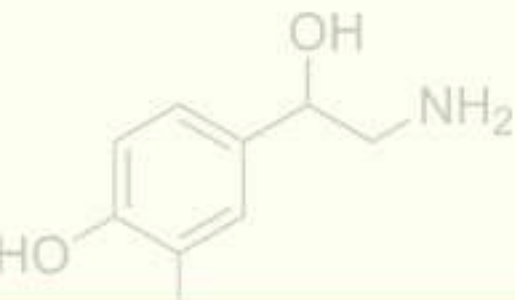


- Salts are essentially ionic compounds .
- Formed by the electrostatic bond between a cation and an anion.
- Salts can also conduct electricity in its molten or aqueous states.
- Salts can have different colours and are used as pigments.
- Salts in our body helps to transmit nerve impulses.





TYPES OF SALTS



WATER OF CRYSTALLIZATION



Water of crystallization or water of hydration is the fixed number of water molecules that occurs inside crystals of one formula unit of a salt.

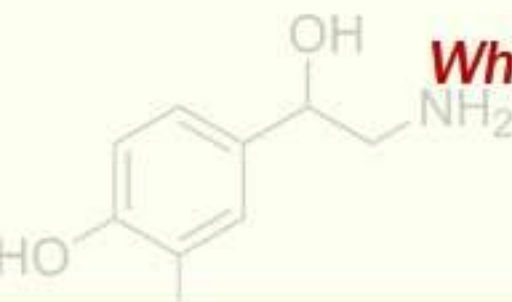
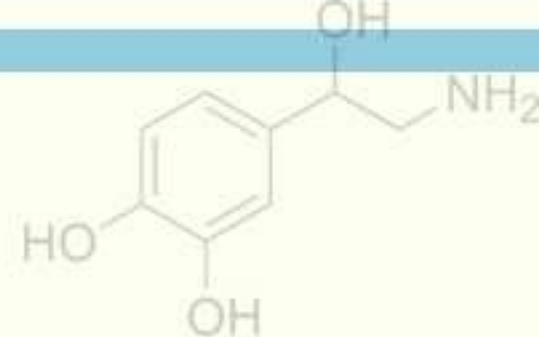
E.G. $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$

$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

When the crystallized salts are heated, they lose the water of crystallization temporarily and become anhydrous



SOME COMMON SALTS

Gypsum



Green Vitriol



Epsom Salt



Blue Vitriol



Mohr's Salt



Washing soda



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